

MAT205: Assignment 1 Solutions

Abstract Algebra II

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Assignment 1

Problem 1

The rotation group of a cube

- body diagonals
- $G \cong S_4$
- Sylow subgroups
- tetrahedral subgroup A_4
- composition factors
- solvable vs. nilpotent

Problem 2

Automorphisms of finite abelian groups

- the group $\mathbb{Z}_{p^2} \oplus \mathbb{Z}_p$
- characteristic filtration
- endomorphism matrices
- automorphism criterion
- primary decomposition

Problem 1: The Cube Rotation Group

Let G be the group of orientation-preserving rotations of a cube.

The cube has four body diagonals:

$$D_1 = \{\pm(1, 1, 1)\}, \quad D_2 = \{\pm(-1, 1, 1)\},$$

$$D_3 = \{\pm(-1, -1, 1)\}, \quad D_4 = \{\pm(1, -1, 1)\}.$$

Goal:

$$G \cong S_4.$$

Body Diagonals Give a Homomorphism

Every rotation sends vertices to vertices and opposite vertices to opposite vertices.

Therefore it sends each body diagonal to another body diagonal.

Define

$$\varphi : G \rightarrow S_4, \quad \varphi(g) = \text{the permutation of } \{D_1, D_2, D_3, D_4\} \text{ induced by } g.$$

For $g, h \in G$, the rotation gh acts by first applying h and then g , so

$$\varphi(gh) = \varphi(g)\varphi(h).$$

Thus φ is a group homomorphism.

Injectivity

Suppose $g \in \ker \varphi$.

Then g fixes each of the four body diagonals as a set.

Nonidentity cube rotations have one of three axis types:

Axis type	Effect on body diagonals
opposite faces	moves all four diagonals
opposite vertices	fixes one diagonal and cycles the other three
opposite edges	swaps diagonals in pairs

No nonidentity rotation fixes all four body diagonals.

Hence $\ker \varphi = 1$, so φ is injective.

Surjectivity: Strategy

It is enough to show every transposition $(ij) \in S_4$ lies in $\text{im } \varphi$.

For a nonzero vector u , the half-turn around the axis $\mathbb{R}u$ is

$$R_u(w) = 2 \frac{w \cdot u}{u \cdot u} u - w.$$

This fixes the line $\mathbb{R}u$ and acts as -1 on $u^\perp$, so its eigenvalues are

$$1, -1, -1.$$

Therefore $\det R_u = 1$, and the half-turn is orientation-preserving.

Six Half-Turns

Use the representatives

$$v_1 = (1, 1, 1), \quad v_2 = (-1, 1, 1), \quad v_3 = (-1, -1, 1), \quad v_4 = (1, -1, 1).$$

Since $D_i = \{\pm v_i\}$, signs do not matter.

axis u	$R_u(x, y, z)$	induced transposition
$(0, 1, 1)$	$(-x, z, y)$	(12)
$(1, 1, 0)$	$(y, x, -z)$	(13)
$(1, 0, 1)$	$(z, -y, x)$	(14)
$(1, 0, -1)$	$(-z, -y, -x)$	(23)
$(1, -1, 0)$	$(-y, -x, -z)$	(24)
$(0, 1, -1)$	$(-x, -z, -y)$	(34)

Conclusion: $G \cong S_4$

The six half-turns put all transpositions in $\text{im } \varphi$.

Since transpositions generate S_4 ,

$$\text{im } \varphi = S_4.$$

We have already proved φ is injective.

Therefore φ is an isomorphism:

$$G \cong S_4.$$

Sylow Counts

Since $G \cong S_4$,

$$|G| = 24 = 2^3 \cdot 3.$$

Sylow gives

$$n_3 \mid 8, \quad n_3 \equiv 1 \pmod{3},$$

so

$$n_3 = 1 \text{ or } 4.$$

Also

$$n_2 \mid 3, \quad n_2 \equiv 1 \pmod{2},$$

so

$$n_2 = 1 \text{ or } 3.$$

Actual Sylow Counts

In S_4 , a Sylow **3**-subgroup is generated by a **3**-cycle.

There are 8 different **3**-cycles, and each subgroup of order **3** contains two nonidentity **3**-cycles.

Thus

$$n_3 = \frac{8}{2} = 4.$$

A Sylow **2**-subgroup has order 8.

Choosing a pair of opposite faces gives the rotations preserving the axis through their centers. These form a subgroup isomorphic to D_4 of order 8.

There are **3** pairs of opposite faces, hence

$$n_2 = 3.$$

Geometric Meaning of the Sylow Subgroups

Sylow 3-subgroup.

Generated by a 120° rotation around a body diagonal.

Geometrically: the rotational symmetry group around a vertex-opposite-vertex axis.

Sylow 2-subgroup.

The order-8 rotation subgroup preserving a chosen pair of opposite faces.

It acts on one square face as the symmetry group generated by quarter-turns and half-turns:

$$D_4, \quad |D_4| = 8.$$

An Inscribed Tetrahedron

Put the cube at the vertices $(\pm 1, \pm 1, \pm 1)$.

Consider

$$T = \{(1, 1, 1), (-1, -1, 1), (-1, 1, -1), (1, -1, -1)\}.$$

The six distances between distinct vertices are all equal.

For example,

$$\|(1, 1, 1) - (-1, -1, 1)\| = \|(1, 1, 1) - (-1, 1, -1)\| = 2\sqrt{2}.$$

Hence T is a regular tetrahedron inscribed in the cube.

The Two Complementary Tetrahedra

The cube vertices split as

$$T = \{(x, y, z) : xyz = 1\}, \quad T' = \{(x, y, z) : xyz = -1\}.$$

Each body diagonal contains exactly one vertex of T and one vertex of T' .

Let

$$K = \{g \in G : g(T) = T\}.$$

The subgroup K consists of the cube rotations that preserve the tetrahedron T setwise.

Why K Corresponds to A_4

The orientation-preserving rotation group of a regular tetrahedron acts on its four vertices as A_4 .

Therefore

$$K \cong A_4.$$

Under the isomorphism

$$G \cong S_4,$$

the subgroup K corresponds to $A_4 \leq S_4$.

Equivalently:

- even permutations preserve each of T and T'
- odd permutations swap the two complementary tetrahedra

Computing V^K

Let $V = \mathbb{R}^3$ and

$$V^K = \{v \in V : k(v) = v \text{ for every } k \in K\}.$$

The subgroup $K \cong A_4$ contains the normal Klein four subgroup

$$V_4 = \{1, (12)(34), (13)(24), (14)(23)\}.$$

Geometrically, this is the identity together with the three 180° rotations around the three face axes.

Their fixed axes are the three coordinate axes, whose intersection is $\{0\}$.

Thus

$$V^K \subseteq V^{V_4} = \{0\}, \quad V^K = \{0\}.$$

Setwise Preservation vs. Pointwise Fixing

The subgroup K preserves the tetrahedron

$$T = \{(1, 1, 1), (-1, -1, 1), (-1, 1, -1), (1, -1, -1)\}$$

as a set.

But it fixes no nonzero vector in $\mathbb{R}^3$ pointwise:

$$V^K = \{0\}.$$

This is the key distinction:

Statement	Meaning
$g(T) = T$	the four vertices may be permuted
$g(v) = v$ for all $g \in K$	one vector is fixed by every element of K

Composition Factors of G

Under $G \cong S_4$, use the chain

$$1 \triangleleft C_2 \triangleleft V_4 \triangleleft A_4 \triangleleft S_4.$$

The successive quotients are

$$C_2/1 \cong \mathbb{Z}_2,$$

$$V_4/C_2 \cong \mathbb{Z}_2,$$

$$A_4/V_4 \cong \mathbb{Z}_3,$$

$$S_4/A_4 \cong \mathbb{Z}_2.$$

Hence the composition factors are

$$\mathbb{Z}_2, \quad \mathbb{Z}_2, \quad \mathbb{Z}_3, \quad \mathbb{Z}_2.$$

Solvable but Not Nilpotent

The group G is **solvable** because it has a subnormal series whose factors are abelian:

$$1 \triangleleft C_2 \triangleleft V_4 \triangleleft A_4 \triangleleft S_4.$$

All factors are cyclic of prime order:

$$\mathbb{Z}_2, \quad \mathbb{Z}_2, \quad \mathbb{Z}_3, \quad \mathbb{Z}_2.$$

The group G is **not nilpotent**.

For a finite group, nilpotent is equivalent to every Sylow subgroup being normal.

But $n_3 = 4$, so the Sylow **3**-subgroups are not unique and therefore not normal.

Problem 2: Automorphisms of Finite Abelian Groups

Let p be a prime and

$$A = \mathbb{Z}_{p^2} \oplus \mathbb{Z}_p.$$

Write

$$A = \langle e_1 \rangle \oplus \langle e_2 \rangle, \quad |e_1| = p^2, \quad |e_2| = p.$$

Goal:

1. compute the natural p -power filtration
2. describe all endomorphisms
3. determine exactly which endomorphisms are automorphisms
4. count $\mathbf{Aut}(A)$

The Natural Subgroups of A

Since e_1 has order p^2 and e_2 has order p ,

$$pA = \langle pe_1 \rangle \cong \mathbb{Z}_p.$$

The elements killed by p are precisely

$$A[p] = \langle pe_1, e_2 \rangle \cong \mathbb{Z}_p \oplus \mathbb{Z}_p.$$

Also

$$A/pA \cong \langle e_1 + pA \rangle \oplus \langle e_2 + pA \rangle \cong \mathbb{Z}_p \oplus \mathbb{Z}_p.$$

The Filtration

Every automorphism of A must preserve

$$0 \subset pA \subset A[p] \subset A.$$

$$A = \langle e_1, e_2 \rangle$$

$$A[p] = \langle pe_1, e_2 \rangle$$

$$pA = \langle pe_1 \rangle$$

$$0$$

Also

$$A/pA \cong \mathbb{Z}_p^2.$$

The filtration records what the group structure forces before any basis choices are made.

Why pA and $A[p]$ Are Characteristic

The subgroup pA is the image of the intrinsic map

$$A \rightarrow A, \quad x \mapsto px.$$

If $\alpha \in \text{Aut}(A)$, then

$$\alpha(px) = p\alpha(x),$$

so $\alpha(pA) = pA$.

Thus pA is characteristic.

The subgroup $A[p]$ is the kernel of the same map.

If $x \in A[p]$, then

$$p\alpha(x) = \alpha(px) = 0,$$

Endomorphism Matrix Form

Let $f : A \rightarrow A$ be an endomorphism.

The image of e_1 can be any element of A , so

$$f(e_1) = ae_1 + be_2,$$

where

$$a \in \mathbb{Z}/p^2\mathbb{Z}, \quad b \in \mathbb{F}_p.$$

Since $pe_2 = 0$, the image $f(e_2)$ must have order dividing p .

Therefore

$$f(e_2) \in A[p] = \langle pe_1, e_2 \rangle,$$

so

$$f(e_2) = pce_1 + de_2, \quad c, d \in \mathbb{F}_p.$$

Matrix Notation

Using columns for the images of e_1, e_2 , every endomorphism has the form

$$\begin{pmatrix} a & pc \\ b & d \end{pmatrix},$$

where

$$a \in \mathbb{Z}/p^2\mathbb{Z}, \quad b, c, d \in \mathbb{F}_p.$$

The entry pc appears because the e_1 -component of $f(e_2)$ must be killed by p .

Necessity for Being an Automorphism

The induced map on A/pA has basis

$$e_1 + pA, \quad e_2 + pA$$

and matrix over $\mathbb{F}_p$

$$\begin{pmatrix} \bar{a} & 0 \\ b & d \end{pmatrix}.$$

If f is an automorphism, this induced map must be invertible.

Hence

$$\bar{a} \neq 0, \quad d \neq 0.$$

Equivalently,

$$a \in (\mathbb{Z}/p^2\mathbb{Z})^\times, \quad d \in \mathbb{F}_p^\times.$$

Sufficiency

Assume

$$a \in (\mathbb{Z}/p^2\mathbb{Z})^\times, \quad d \in \mathbb{F}_p^\times.$$

Suppose $f(x) = 0$ and write

$$x = ue_1 + ve_2.$$

Reducing modulo pA gives

$$\bar{a}\bar{u} = 0, \quad b\bar{u} + d\bar{v} = 0$$

in $\mathbb{F}_p$.

Since $\bar{a} \neq 0$ and $d \neq 0$, we get

$$\bar{u} = 0, \quad v = 0.$$

Thus $x = u'pe_1 \in pA$.

Sufficiency: Finish

On pA , the map sends

$$pe_1 \mapsto pf(e_1) = p(ae_1 + be_2) = a pe_1.$$

Since a is a unit modulo p , this is an automorphism of pA .

Therefore $x = 0$.

So f is injective, and because A is finite, f is an automorphism.

Hence

$$f \in \text{Aut}(A) \iff a \in (\mathbb{Z}/p^2\mathbb{Z})^\times, \quad d \in \mathbb{F}_p^\times.$$

Counting $\text{Aut}(A)$

An automorphism is determined by

$$\begin{pmatrix} a & pc \\ b & d \end{pmatrix},$$

where

$$a \in (\mathbb{Z}/p^2\mathbb{Z})^\times, \quad b, c \in \mathbb{F}_p, \quad d \in \mathbb{F}_p^\times.$$

There are

$$|(\mathbb{Z}/p^2\mathbb{Z})^\times| = p^2 - p = p(p-1)$$

choices for a , p choices for each of b, c , and $p-1$ choices for d .

Thus

$$|\text{Aut}(A)| = p(p-1) \cdot p \cdot p \cdot (p-1) = p^3(p-1)^2.$$

General Finite Abelian p -Groups

Let

$$B = \mathbb{Z}_{p^{\lambda_1}} \oplus \cdots \oplus \mathbb{Z}_{p^{\lambda_r}}, \quad \lambda_1 \geq \cdots \geq \lambda_r.$$

For any $k \geq 0$, the subgroup $p^k B$ is the image of the intrinsic map

$$B \rightarrow B, \quad x \mapsto p^k x.$$

Every automorphism commutes with multiplication by p^k .

Therefore every $p^k B$ is characteristic.

The General Principle

A finite abelian p -group carries a natural p -power filtration:

$$B \supset pB \supset p^2B \supset \cdots .$$

Automorphisms must preserve this filtration.

In a basis adapted to the cyclic summands, an endomorphism is represented by a matrix whose entries satisfy divisibility conditions forced by the orders of the generators.

It is an automorphism exactly when the induced maps on the relevant finite layers are invertible.

Primary Decomposition

Let M be any finite abelian group with primary decomposition

$$M = \bigoplus_p M_p, \quad M_p = \{m \in M : p^k m = 0 \text{ for some } k \geq 1\}.$$

Each M_p is characteristic.

Indeed, if $\alpha \in \text{Aut}(M)$ and $m \in M_p$, then $p^k m = 0$ for some k , so

$$p^k \alpha(m) = \alpha(p^k m) = 0.$$

Thus $\alpha(m) \in M_p$.

Automorphisms Split by Prime

Because every automorphism preserves each primary component, restriction gives

$$\text{Aut}(M) \rightarrow \prod_p \text{Aut}(M_p).$$

This map is injective because an automorphism of

$$M = \bigoplus_p M_p$$

is determined by its restrictions to the direct summands.

It is surjective because any tuple

$$(\alpha_p)_p, \quad \alpha_p \in \text{Aut}(M_p),$$

defines an automorphism of M by acting as α_p on the p -primary component.

Therefore

Assignment 1: Main Takeaways

Problem 1:

$$\text{Rot}^+(\text{cube}) \cong S_4.$$

The geometry of the cube gives:

- Sylow subgroup counts and meanings
- the tetrahedral subgroup A_4
- composition factors
- solvability but not nilpotence

Problem 2:

- pA , $A[p]$, and A/pA are forced by the group structure
- automorphisms preserve the p -power filtration
- for $A = \mathbb{Z}_{p^2} \oplus \mathbb{Z}_p$,